

QueueFlow: Computer Vision Based Automated Queue Ticket Assignment and Real-Time Predictive System for Dynamic Campus Service Delivery

A Capstone Research

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Naga College Foundation, Inc.
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The Requirements for the Degree
Bachelor of Science in Computer Science

Balbin, Archie D.
Sanota, Joshua Rovic P.

NAGA COLLEGE FOUNDATION, INC.
College of Computer Studies
City of Naga

APPROVAL SHEET

This capstone research of **ARCHIE D. BALBIN** and **JOSHUA ROVIC P. SAN-OTA** entitled “**QUEUEFLOW: COMPUTER VISION BASED AUTOMATED QUEUE TICKET ASSIGNMENT AND REAL-TIME PREDICTIVE SYSTEM FOR DYNAMIC CAMPUS SERVICE DELIVERY**” was prepared and submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science. It has been examined and is recommended for acceptance and approval for oral examination.

[ADVISER NAME]
Adviser

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Panel Member

[PANEL MEMBER 2]
Panel Member

[SECRETARY NAME]
Panel Secretary

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[CHAIRMAN NAME]

Chairman

[PANEL MEMBER 1]

Panel Member

[PANEL MEMBER 2]

Panel Member

Accepted and approved in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science.

ANNA LORETTA C. ROMULO, MSIT

Dean

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SECRETARY'S CERTIFICATION

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[**SECRETARY NAME**]
Secretary

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City of Naga

EDITOR'S CERTIFICATION

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[EDITOR NAME]
Editor

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Abstract

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Queue management in Philippine higher education institutions remains predominantly manual, resulting in inefficient counter assignments, inconsistent service monitoring, and prolonged student wait times during peak periods. This paper presents QueueFlow, a Smart Queue Management System that uses You Only Look Once version 8 nano (YOLOv8n) with object tracking, OpenCV camera processing, automated queue-number assignment, ticket generation, and real-time wait-time prediction. The implemented prototype separates the detector from the web backend: a camera-connected detector process performs YOLO inference, filters duplicate and implausible detections, estimates crowd metrics, and pushes tracked-person metadata to a FastAPI backend. The backend confirms new queue entrants through a configurable queue zone, candidate confirmation window, motion checks, track-ID remapping, and hybrid appearance-spatial re-identification before issuing a queue number. The intended deployment issues physical queue tickets through a thermal printer; for prototype demonstration, the system temporarily generates PDF tickets containing the same queue number, QR code, short code, and JSON Web Token (JWT). Ticket records are stored in MySQL, and students can check live status through a mobile browser or mobile tracker. The staff dashboard displays live annotated snapshots, queue state, no-show alerts, counter controls, analytics, and forecasts. Prediction outputs include current estimated wait time and 5, 15, and 30 min forecasts using an M/M/c wait-time baseline, short-term trend projection, Holt's double exponential smoothing, and a mean-reversion long forecast. Developed for Naga College Foundation, Inc. (NCF) in Naga City, Philippines, QueueFlow is evaluated in terms of queue assignment accuracy, re-identification reliability, ticket and token validation,

prediction accuracy using MAE and MAPE, system latency, and user satisfaction.

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Chapter 1

Introduction

1.1 Background of the Study

Campus service delivery depends not only on the availability of staff and counters, but also on how quickly an office can recognize demand as it forms. During enrollment, payment, clearance, and other academic transactions, students often gather faster than service personnel can manually count, organize, and update the queue. This creates a practical problem for both sides of the service process: students have limited visibility of their waiting status, while staff have limited real-time information for deciding when to serve, skip, recall, or prepare additional service capacity.

At Naga College Foundation Inc. (NCF), queue congestion is not simply a matter of long lines. It affects the continuity of student transactions because campus offices such as enrollment, cashiering, and library services are part of a connected workflow. A delay in one service point can influence the next step a student needs to complete. The Commission on Higher Education recognizes accessibility and efficient service delivery as important indicators of institutional quality [1]; therefore, improving queue visibility is also part of improving the overall student service experience.

Conventional queue systems help arrange people in order, but they usually begin only after a student manually asks for a number or after staff encode the transaction. These systems can record issued tickets, yet they do not automatically observe the actual service area, confirm new arrivals from a camera view, prevent duplicate queue entries, or estimate how the line may change in the next few minutes. Earlier queue management tools also tend to focus on static numbering rather than combining automated detection, staff monitoring, student status access, and prediction in one workflow.

QueueFlow is proposed in this study as a computer vision based queue management prototype for campus service offices. Instead of relying only on manual queue entry, the system uses a camera-connected YOLOv8n and OpenCV pipeline to detect persons inside a defined queue zone. YOLO is suitable for this task because it performs real-time object detection in a single network pass [2], and YOLOv8n provides a lightweight model that can be used in practical prototype settings [3]. QueueFlow extends detection with duplicate filtering, candidate confirmation, and hybrid re-entry matching so that queue records remain more consistent when people move or briefly leave the camera frame.

The queue ticket component of QueueFlow is designed for thermal printer deployment, allowing a physical queue ticket to be issued after a queue number is assigned. Because the prototype is demonstrated without the final printer hardware, the current implementation generates a PDF ticket as a temporary substitute. This PDF carries the same operational information expected from the thermal ticket, including the queue number, short code, and QR link for student status checking.

Beyond ticket generation, QueueFlow connects the detector, queue records, staff dashboard, and prediction module through a FastAPI-based backend and MySQL database. The prediction module supports current, 5-minute, 15-minute, and 30-minute wait-time estimates using queuing logic and short-term forecasting. Through this design, the study aims to provide NCF with a prototype that improves queue visibility, supports staff decision-making, and demonstrates how computer vision and predictive analytics can be applied to campus service management in a Philippine educational institution.

The Review of Related Literature therefore examines the technologies and concepts that support each part of the problem identified above. Computer vision and YOLO literature address automated queue observation, queue management literature explains ticketing and identity maintenance, and forecasting literature supports the wait-time estimates needed for staff decision-making.

1.2 Review of Related Literature

Computer Vision for Real-Time Person Detection. Computer vision has matured significantly through deep convolutional neural networks, enabling non-intrusive observation of crowd density and movement in public service areas. Sindagi and Patel [4] surveyed CNN-based approaches to single-image crowd counting and density estimation, documenting how deep learning has transformed earlier methods limited to small variations in scale and density into systems capable of handling occlusions, complex scenes, and dense crowds. One-stage detectors such as YOLO have enabled live deployment at frame rates suitable for real-time monitoring [2]. A comprehensive review by Terven et al. [5] traces YOLO’s evolution from YOLOv1 through YOLOv8 and YOLO-NAS, examining the architectural innovations that drive each iteration’s accuracy-speed trade-off. These advances inform QueueFlow’s design, which processes camera input through a YOLOv8n pipeline rather than relying on manual observation of queue activity.

Privacy concerns are an essential consideration in any camera-based system. Padilla-Lopez et al. [6] survey visual privacy protection methods, categorizing approaches such as anonymization at the camera level, selective scrambling, and aggregate-only transmission. These principles inform QueueFlow’s design choice to process bounding-box data and colour histograms without storing or transmitting raw video.

YOLO-Based Tracking and Deduplication. YOLO’s single-pass detection architecture delivers the speed necessary for real-time queue monitoring, processing entire frames in one forward pass to simultaneously predict bounding boxes, confidence scores, and class labels [2]. The YOLOv8 series advances this through improved multi-scale feature pyramid networks and anchor-free detection heads [3], with the broader YOLO family reviewed in depth by Terven et al. [5].

Beyond single-frame detection, multi-object tracking requires associating detections across frames into consistent identities. Wojke et al. [7] introduced the DeepSORT algorithm, which integrates appearance information with motion-based tracking via Kalman filtering and the Hungarian algorithm, reducing identity switches when targets are temporarily occluded. QueueFlow draws on similar principles in its custom track-ID remapping logic, while adding a domain-specific NMS pre-filter using X-column overlap and Y-adjacency thresholds to suppress the duplicate head/body boxes YOLO commonly emits for persons at close camera range.

Queue Management and Person Re-Identification. Camera-based monitoring of service areas enables passive observation of arrivals without requiring user interaction. The core technical challenge beyond detection is person re-identification: when a tracked individual temporarily disappears from view and reappears, the system must recognize them as the same person and restore their queue position rather than assigning a new number.

Ye et al. [8] provide a comprehensive survey of deep learning approaches to person re-identification, dividing methods into closed-world and open-world settings and categorizing the field along three axes: deep feature representation learning, deep metric learning, and ranking optimization. Their analysis underscores that practical re-ID systems typically combine multiple signals: appearance-based methods using colour histograms or deep feature embeddings, spatial-temporal methods using last known position and elapsed time, and hybrid methods combining both. QueueFlow implements a hybrid re-entry matching approach in this tradition: it first matches a new detection to a missing person by spatial proximity, then by appearance histogram similarity, and finally by recency as a last resort.

Predictive Analytics for Service Demand Forecasting. Multi-horizon forecasting provides both immediate and strategic insights by generating predictions at different time scales, each matched to a specific operational decision type. Shmueli and Koppius [9] formalize predictive analytics as a research paradigm distinct from explanatory modelling, emphasizing the role of empirical methods that generate and assess data predictions for decision support.

The M/M/c queuing model provides analytical wait-time calculations from real-time arrival and service rate measurements, making it suitable for current wait-time estimates and near-term tactical predictions [10]. Together with Little’s Law, the M/M/c framework gives QueueFlow the mathematical basis for translating live arrival rates into expected waiting times. System utilization is expressed as:

$$\rho = \frac{\lambda}{c\mu} \tag{1.1}$$

where λ is the arrival rate, μ the service rate, and c the number of active servers. The M/M/c model processes real-time arrival rates from computer vision systems to calculate immediate wait times using Little’s Law:

$$W = \frac{L}{\lambda} \tag{1.2}$$

For medium-term forecasting, Holt’s double exponential smoothing tracks both queue

level and trend with separate smoothing parameters [11]. Hyndman and Athanasopoulos [12] provide a comprehensive textbook treatment of exponential smoothing methods and multi-horizon forecasting. Damping the trend component prevents over-projection during transient spikes, a practical necessity in campus service environments where queue lengths can surge and dissipate quickly. For the 30 min forecast implemented in QueueFlow, mean-reversion logic is used to limit long-forecast uncertainty.

1.3 Synthesis of the State-of-the-Art

The literature reviewed in this chapter supports the central concern identified in the Background of the Study: campus service offices need better real-time visibility of queue conditions, not merely a list of issued numbers. Studies on computer vision show that camera-based person detection can reduce dependence on manual observation by capturing activity in a service area as it happens. YOLO-based approaches are especially relevant because they provide the speed needed for live monitoring, making them suitable for a prototype that must observe students entering and moving through a defined queue zone.

However, the reviewed studies also show that detection is only the first layer of a complete queue management system. A person detector can identify bounding boxes, but it does not automatically solve the operational problems described in the Background: duplicate entries, temporary disappearance from the camera view, skipped queue positions, or student access to queue status. This makes tracking, deduplication, candidate confirmation, and re-entry matching necessary additions when computer vision is applied to actual service-office queues.

Research on queue management and predictive analytics further reinforces the need to move beyond static ticket numbering. Traditional queue tools can arrange students in order, but they provide limited support for staff decisions when demand changes quickly. Queuing theory and short-term forecasting address this limitation by translating arrival rates, active counters, and queue length into wait-time estimates. Multi-horizon prediction is therefore useful because staff need both immediate queue status and near-future indicators for counter planning.

Taken together, the reviewed works suggest that an effective campus queue solution should combine four functions in one workflow: camera-based queue entry, reliable queue identity management, ticket and status access for students, and predictive information for staff. These findings connect directly to QueueFlow’s design, which integrates YOLOv8n detection, hybrid re-entry matching, thermal-ticket-oriented queue number generation with a PDF demonstration fallback, a staff dashboard, and current, 5-minute, 15-minute, and 30-minute wait-time estimates.

Despite these advances, existing studies remain limited when viewed against the specific needs of NCF’s campus service context. Many focus on crowd counting, general queue applications, or prediction models as separate components. Fewer works demonstrate how these components can operate together in a Philippine educational institution where queue visibility, staff action, student status checking, and practical deployment constraints must be handled in a single prototype.

1.4 Gap Bridged by the Study

This study addresses a specific practical gap identified from the literature reviewed in Section 1.2. While existing studies discuss queue monitoring, computer vision, person detection, re-identification, and wait-time prediction, they do not report an integrated system that combines passive camera-based queue entry using YOLO, hybrid multi-tier person re-identification, and multi-algorithm, multi-horizon wait-time prediction specifically designed for campus service delivery in a Philippine higher education institution context.

QueueFlow bridges this gap by developing these components within a unified system architecture. It implements a YOLO-based person detection and tracking pipeline supported by a custom non-maximum suppression pre-filter and post-frame deduplication layer to improve queue entrant detection. It also applies a hybrid re-entry matching algorithm that combines spatial proximity, appearance similarity, and recency to reduce duplicate queue assignments when a person temporarily leaves and returns to the queue area.

In addition, the system supports thermal-ticket-oriented queue number generation, with PDF ticket generation used as a demonstration fallback. QueueFlow also exposes wait-time prediction outputs for the current queue condition and for 5-minute, 15-minute, and 30-minute horizons. Through this integration, the study bridges the gap between existing queue management research and the need for a practical, automated, campus-oriented queue monitoring system for students and administrative staff.

1.5 Theoretical Framework

The theoretical framework of this study is anchored on the queue visibility problem described in the Background of the Study. Campus service offices need a way to observe student arrival, maintain queue order, estimate waiting time, and present useful information to both staff and students. QueueFlow addresses this problem through the combined application of queuing theory, computer vision, predictive analytics, and forecasting theory.

Queuing Theory, as formalized in classical queueing literature [10], explains how arrivals, service capacity, and waiting lines behave through the M/M/c model. The M/M/c model represents a multi-counter service environment in which the arrival rate, service rate, and number of active counters influence queue length, utilization, and waiting time. Together with Little’s Law, this theory provides the mathematical basis for estimating waiting time, queue length, and counter utilization from real-time arrival and service rates. In QueueFlow, queuing theory is connected to the system through queue records, active counter settings, and service-time assumptions.

Computer Vision Theory provides the basis for replacing manual line observation with automated person detection. In QueueFlow, this theory is applied through YOLOv8n and OpenCV, where camera frames are processed to detect persons inside a defined queue zone. YOLO’s single-pass detection approach supports real-time monitoring because it can identify objects quickly within each frame [2]. Computer vision supplies the first

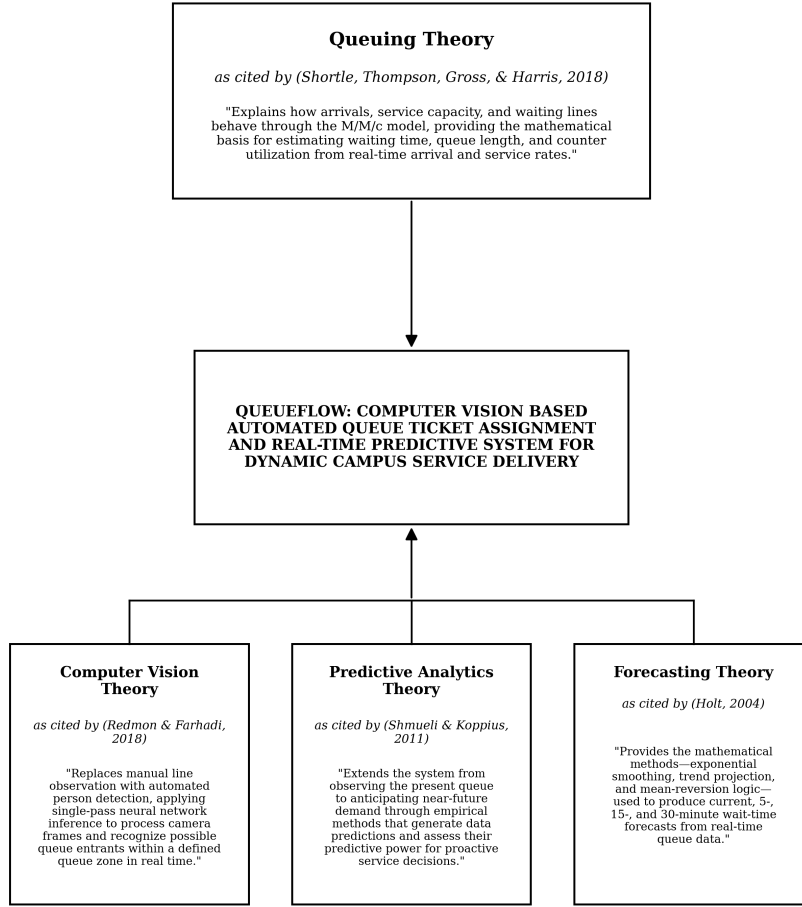


Figure 1: Theoretical Paradigm

Figure 1.1: Theoretical Framework of QueueFlow

layer of queue visibility: it allows the system to recognize possible queue entrants without requiring every student to manually register at a counter.

Predictive Analytics Theory extends the system from observing the present queue to anticipating near-future demand. As articulated in information systems research, predictive analytics comprises empirical methods that generate data predictions and assess their predictive power for decision support [9]. In QueueFlow, predictive analytics is the conceptual basis for transforming real-time arrival data into proactive staff and counter management.

Forecasting Theory supplies the specific mathematical methods used for the prediction module. QueueFlow applies exponential smoothing, trend projection, and mean-reversion logic—methods rooted in the classical exponentially weighted moving-average framework [11] and treated comprehensively by Hyndman and Athanasopoulos [12]—to produce current, 5-minute, 15-minute, and 30-minute wait-time estimates. These fore-

casts support staff decision-making by giving the dashboard more than a static queue count; they provide an early indication of whether congestion is likely to increase or decrease.

Together, these theories form the foundation of QueueFlow’s design. Computer vision converts camera input into observable queue activity; queuing theory converts queue activity and counter capacity into service metrics; predictive analytics provides the conceptual framework for anticipatory decision-making; and forecasting theory supplies the mathematical methods that turn metrics into near-future wait-time estimates. The resulting information is delivered through the staff dashboard, student status access, and queue ticket workflow, where the intended deployment uses a thermal printer and the current prototype uses PDF tickets as a demonstration fallback.

1.6 Conceptual Framework

This study uses the Input-Process-Output-Outcome (IPOO) model as the conceptual framework for developing QueueFlow. The IPOO framework demonstrates how camera-based queue observations, system configuration, staff actions, and service assumptions are transformed through a computer-vision and prediction pipeline into functional outputs that support the intended queue management outcomes.

Figure 1.2 presents the IPOO-based conceptual framework of the QueueFlow system, showing the relationship among inputs, processes, outputs, outcomes, and feedback.

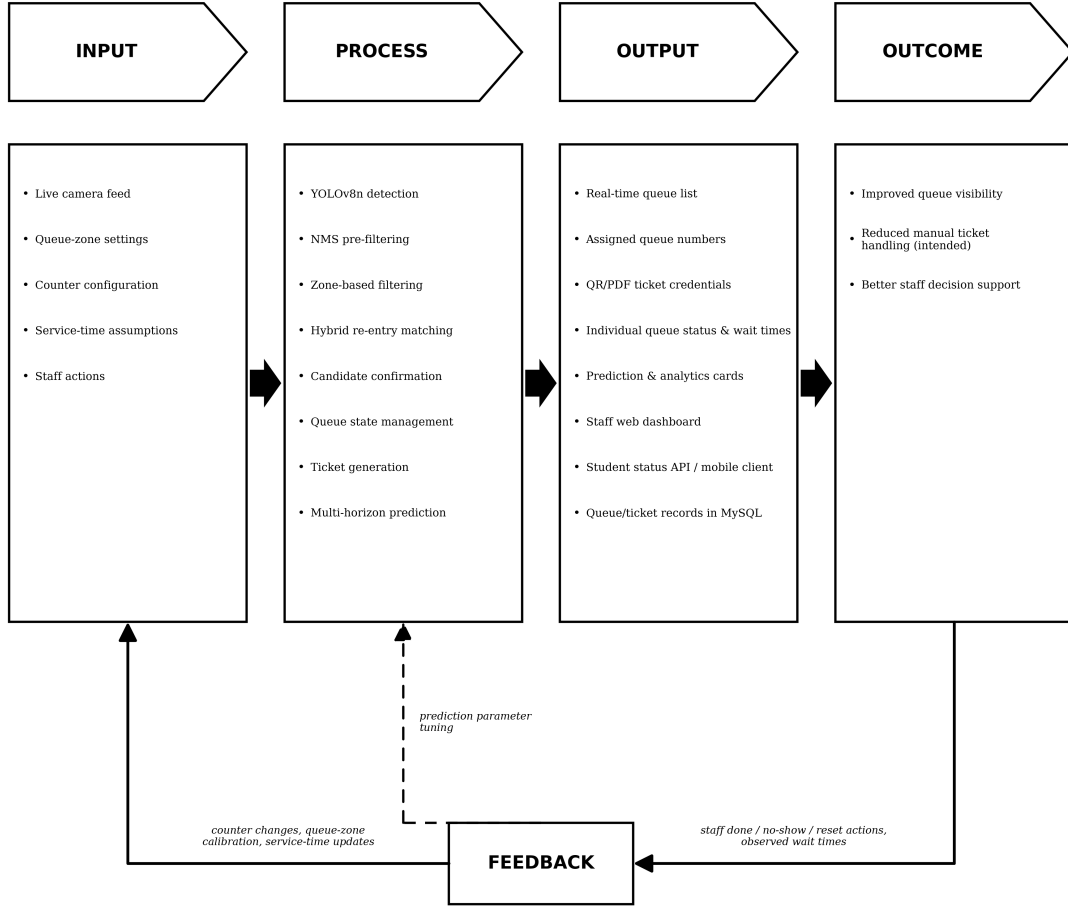


Figure 1.2: Conceptual Framework of QueueFlow

Input. The initial phase consists of the data, settings, and operational conditions required by the system. These include the live camera feed, queue-zone settings, counter configuration, service-time assumptions, staff actions, and the ticket and status-access requirements used by students and administrative staff.

Process. The process stage transforms the input data into queue intelligence through the QueueFlow pipeline. Key processes include YOLOv8n person detection, NMS pre-filtering, zone-based filtering, hybrid re-entry matching, candidate accumulation over a minimum confirmation window, queue state management, no-show detection, position recalculation, ticket generation, and multi-horizon wait-time prediction.

Output. The main outputs are a real-time queue list with assigned queue numbers, QR/PDF ticket credentials, individual queue status and wait times, prediction and analytics cards, the staff web dashboard, and the student status API consumed by a mobile client. Queue entries, ticket records, and operational events are also persisted in the

MySQL database.

Outcome. The expected outcome is improved queue visibility and better staff decision support. Reduced manual ticket handling is also an intended outcome of the full thermal-printer deployment.

Feedback. Observed wait times together with staff done, no-show, and reset actions are routed into the Feedback stage. This feedback propagates counter changes, queue-zone calibration, and measured service-time updates back into the Input stage. A secondary signal, shown as a dashed arrow in Figure 1.2, feeds prediction-parameter tuning back into the Process stage, allowing the candidate confirmation window, smoothing factors, and mean-reversion thresholds to be adjusted iteratively. This refinement allows QueueFlow to adapt to the actual demand patterns of NCF service offices rather than remaining locked to its initial settings.

1.7 Statement of the Problem

This study seeks to address the following problems:

1. What are the current limitations of the manual queue management process at NCF service offices during peak demand periods?
2. How effectively can a YOLO-based passive detection pipeline with hybrid re-identification assign and maintain queue numbers without student interaction, as measured by queue assignment accuracy, re-identification reliability, ticket and token validation, and detection-to-ticket latency?
3. To what degree of accuracy, measured by Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE), can the implemented prediction models forecast current, 5-, 15-, and 30-minute wait times?
4. How do service staff and administrators evaluate QueueFlow in terms of usability, detection reliability, prediction usefulness, dashboard clarity, and overall satisfaction?

1.8 Assumptions

This study is based on the following assumptions:

Camera visibility and queue-zone coverage. The camera hardware provides sufficient video quality and coverage of the queue zone to enable reliable YOLO person detection under the lighting conditions and pedestrian densities typical of NCF service areas.

Effectiveness of tracking and re-identification. YOLO’s tracking mechanism, supplemented by the system’s spatial re-identification layer, produces sufficiently stable person identities across the queue duration.

Usefulness of prediction outputs. The implemented prediction models generate forecasts that are actionable and sufficiently accurate for staff scheduling and counter management decisions.

Ticket generation and printer deployment. The thermal printer can be configured in the deployment area. Where printer hardware is unavailable, PDF ticket generation can demonstrate the same queue-number, QR-code, and token workflow.

Staff capability and system adoption. Service staff and administrators possess basic digital literacy to operate the web dashboard and interpret the system’s outputs.

1.9 Scope and Delimitation

This study centers on the design, implementation, and evaluation of QueueFlow at NCF in Naga City, Philippines. The scope covers the development of a YOLOv8n-based detection pipeline, a hybrid re-entry matching algorithm, a multi-horizon prediction engine, a FastAPI-based REST API, a MySQL-backed data layer, and a ticket generation module designed for thermal-printer deployment. The current demonstration build generates PDF tickets with QR codes as a temporary substitute for printed thermal tickets.

The study is limited to a single-queue-zone, single-service prototype deployment. It does not currently support simultaneous multi-zone or multi-department management, priority queue differentiation based on student status or urgency factors, direct integration with the Campus Management System or Student Information System, or IoT sensor integration beyond the camera feed and queue ticket printer. The evaluation period is restricted to one semester of post-implementation operation.

1.10 Significance of the Study

This study examines the development and potential impact of QueueFlow as a computer-vision-based queue monitoring and ticketing system for NCF service offices, highlighting its significance to the following stakeholders:

Service Staff. Service staff gain a real-time queue dashboard that reduces manual counting and provides data-driven insight into current queue length, individual wait times, and demand forecasts across multiple time horizons.

Students. Students benefit from a passive queueing process that does not require manual registration. Upon entering the monitored area, a queue number is assigned and can be checked through the status link or mobile client using the ticket credential.

Institutional Administrators. Administrators gain queue performance records that support evidence-based decisions about staffing, service hours, and physical layout.

Naga College Foundation, Inc. The institution benefits from a campus-oriented prototype that demonstrates how automation, computer vision, and predictive analytics can support service delivery and student-facing operations.

Academic Community and Future Researchers. The study advances the integration of computer vision, person re-identification, ticket generation, and multi-algorithm predictive analytics in service queue management, providing a foundation for related research and system improvements.

1.11 Definition of Terms

Access Token. A ticket credential assigned to each confirmed queue number. In the implemented prototype, this consists of a short code and a JWT-backed validation record that allows the student to check queue status without a login account.

Candidate Accumulation. A buffer mechanism requiring a new detection to appear consistently for a minimum confirmation window before a queue number is assigned, preventing transient detections from generating spurious tickets. The prototype uses a configurable default of 14 confirmation frames.

Holt’s Double Exponential Smoothing. A classical forecasting algorithm [11] used for the 15 min prediction, applying separate smoothing parameters for queue level ($\alpha = 0.4$) and trend ($\beta = 0.2$) with damping factor 0.85 to avoid over-projection.

M/M/c Model. A foundational multi-server queueing model from classical queueing theory [10] used for short-term (5–15 min) wait time prediction, calculating $W = L/\lambda$ from Little’s Law using real-time arrival rates from the YOLO detection pipeline.

Non-Maximum Suppression (NMS). Both a standard IoU-based NMS pass and a custom X-column/Y-adjacency filter applied to raw YOLO outputs before they reach the queue tracker, suppressing duplicate head and body boxes YOLO commonly emits for a single person.

Queue Zone. A configurable rectangular region defined by four pixel coordinates on the camera frame; only persons whose bounding box centre falls within this region are tracked and assigned queue numbers.

Thermal Queue Ticket. The intended physical output of the system containing the queue number, QR code, and short code. In the prototype demonstration, the same ticket information is generated as a PDF while thermal printer hardware is not yet attached.

Re-Entry Matching. A hybrid algorithm that first matches a new detection to a missing person by spatial proximity, then by appearance histogram similarity, and finally by recency as a last resort.

You Only Look Once (YOLO). The YOLOv8n model [3] with `persist=True` tracking, configurable input size, and confidence threshold, used to detect and track persons entering the queue zone in camera frames. A comprehensive review of the YOLO family is provided by Terven et al. [5].

Chapter 2

Review of Related Literature

2.1 Computer Vision for Real-Time Person Detection

Computer vision has matured significantly through deep convolutional neural networks, enabling non-intrusive observation of crowd density and movement in public service areas. Sindagi and Patel [4] surveyed CNN-based approaches to single-image crowd counting and density estimation, documenting how deep learning has transformed earlier methods limited to small variations in scale and density into systems capable of handling occlusions, complex scenes, and dense crowds. One-stage detectors such as YOLO have enabled live deployment at frame rates suitable for real-time monitoring [2]. A comprehensive review by Terven et al. [5] traces YOLO’s evolution from YOLOv1 through YOLOv8 and YOLO-NAS, examining the architectural innovations that drive each iteration’s accuracy-speed tradeoff. These advances inform QueueFlow’s design, which processes camera input through a YOLOv8n pipeline rather than relying on manual observation of queue activity.

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2.2 YOLO-Based Tracking and Deduplication

YOLO’s single-pass detection architecture delivers the speed necessary for real-time queue monitoring, processing entire frames in one forward pass to simultaneously predict bounding boxes, confidence scores, and class labels [2]. The YOLOv8 series advances this through improved multi-scale feature pyramid networks and anchor-free detection heads [3], with the broader YOLO family reviewed in depth by Terven et al. [5].

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2.3 Queue Management and Person Re-Identification

Camera-based monitoring of service areas enables passive observation of arrivals without requiring user interaction. The core technical challenge beyond detection is person re-identification: when a tracked individual temporarily disappears from view and reappears, the system must recognize them as the same person and restore their queue position rather than assigning a new number.

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2.4 Predictive Analytics for Service Demand Forecasting

Multi-horizon forecasting provides both immediate and strategic insights by generating predictions at different time scales, each matched to a specific operational decision type. Shmueli and Koppius [9] formalize predictive analytics as a research paradigm distinct from explanatory modelling, emphasizing the role of empirical methods that generate and assess data predictions for decision support.

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can surge and dissipate quickly. For the 30 min forecast implemented in QueueFlow, mean-reversion logic is used to limit long-forecast uncertainty.

Chapter 3

Methodology

3.1 Research Design

This study utilized a Mixed-Methods and Developmental research design to systematically develop and evaluate QueueFlow. Quantitative data focused on technical performance metrics such as queue number assignment accuracy, re-entry matching accuracy, prediction model MAE and MAPE across three forecasting horizons, and real-time processing latency. Qualitative data was gathered through semi-structured interviews exploring the experiences and perceptions of service staff and administrators.

3.2 Population and Locale

The study was conducted at Naga College Foundation, Inc. (NCF) located in Naga City, Camarines Sur, Philippines. NCF was selected as the research site because it serves large volumes of student service transactions across enrollment, cashiering, and library operations, with pronounced peak periods that create the queue management challenges QueueFlow is designed to address.

3.3 Research Instrument

The study utilized a researcher-made structured questionnaire employing a five-point Likert scale to measure respondents' level of agreement with statements related to system usability, detection reliability, prediction usefulness, dashboard clarity, and overall satisfaction as shown in Table 3.1.

3.4 Ethical Considerations

This study strictly adheres to ethical standards to protect participants' rights, ensure data privacy, and maintain research integrity. Participation was entirely voluntary and informed consent was obtained from all participants. All personal information was

Table 3.1: Likert Scale Interpretation

Scale	Category	WM Range	Interpretation
5	Strongly Agree	4.21–5.00	Excellent
4	Agree	3.41–4.20	Very Satisfactory
3	Neutral	2.61–3.40	Satisfactory
2	Disagree	1.81–2.60	Poor
1	Strongly Disagree	1.00–1.80	Very Poor

anonymized and stored in password-protected systems. The study complied with the Data Privacy Act of 2012 (Republic Act No. 10173).

3.5 Statistical Treatment of Data

Data gathered was processed using both descriptive and technical performance metrics. Descriptive analysis used the Weighted Mean ($WM = \Sigma(f \times w)/N$), Overall Weighted Mean, and Percentage to summarize survey responses. System performance was evaluated using:

$$MAE = \frac{\sum |y_{\text{predicted}} - y_{\text{actual}}|}{n} \quad (3.1)$$

$$MAPE = \frac{1}{n} \sum \left| \frac{y_{\text{actual}} - y_{\text{predicted}}}{y_{\text{actual}}} \right| \times 100\% \quad (3.2)$$

calculated separately for each of the three prediction model horizons.

3.6 System Design and Architecture

QueueFlow is implemented as a FastAPI-based web application with a separate camera detector process, a MySQL backend, staff dashboard pages, and public queue-status endpoints for students as shown in Table 3.2. The detector process captures frames through OpenCV `VideoCapture`, runs YOLOv8n tracking with configurable image size and confidence threshold, filters duplicate and implausible bounding boxes, extracts crowd and prediction metrics, and sends tracked-person metadata to the backend. The backend then performs queue-zone filtering, candidate confirmation, track-ID remapping, re-entry matching, queue-state management, and ticket generation. The intended ticket output is a thermal queue ticket; the demonstration build uses generated PDF tickets with QR codes as a temporary substitute.

3.7 Software Development Process

The system was developed iteratively using the Agile development methodology, organized into sprint-based cycles covering: YOLO pipeline setup and NMS pre-filter development; re-entry matching algorithm design and debugging; no-show detection imple-

Table 3.2: System Tools and Technologies

Category	Technology	Rationale
Detection	YOLOv8n	Lightweight person detection with persistent tracking
Backend	FastAPI	REST API, dashboard routes, detector upload endpoints, and MJPEG streaming
Video	OpenCV (cv2)	Camera capture, frame preprocessing, annotation, and JPEG encoding
Tracking	Custom Python	Queue-zone filtering, track remapping, re-entry matching, and no-show handling
Prediction	Custom Python	M/M/c baseline, short trend projection, Holt smoothing, and mean reversion
Ticketing	Thermal printer; demo PDF	Physical queue ticket output, with PDF/QR substitute for demonstration
Database	MySQL	Queue records, short codes, JWTs, events, and transaction history
Auth	Session/JWT	Staff session login, camera-token validation, and student ticket tokens

mentation; prediction model development and accuracy tuning; FastAPI REST API and staff dashboard development; thermal-ticket workflow design with temporary PDF ticket generation for demonstration; and student queue-status integration testing.

3.8 Validation and Testing Methods

System validation was conducted through four primary approaches. *Alpha Testing* was conducted internally by the developers to identify and resolve functional issues. *Beta Testing* released the system to a limited group of service staff at NCF to observe real-world interaction and collect feedback. *Usability Evaluation* employed the structured Likert-scale questionnaire administered to service staff and administrators after a minimum of two weeks of regular system use. *Performance Testing* evaluated processing latency, prediction accuracy using MAE and MAPE against ground-truth wait times, re-entry matching accuracy verified against direct observation logs, and ticket flow validation from queue assignment to QR/short-code status lookup.

Chapter 4

Synthesis Matrix of Related Literature

Table 4.1 presents a synthesis matrix summarizing the reviewed literature, the methodology of each study, key findings, and the gaps identified that motivated the present work.

Table 4.1: Synthesis Matrix of Related Literature

Author/s	Year	Topic/Focus	Findings	Gap/Limitation
Sindagi & Patel [4]	2018	CNN-based crowd counting and density estimation	Deep learning enables robust counting under occlusion and scale variation	No queue-number assignment or re-identification
Redmon & Farhadi [2]	2018	YOLOv3 real-time object detection	Single-pass network predicts bounding boxes and class probabilities	No queue-specific deduplication
Jocher et al. [3]	2023	YOLOv8 multi-scale anchor-free detection	Accuracy-speed tradeoff with persistent tracking support	Duplicate head/body boxes in close-range scenarios
Terven et al. [5]	2023	Comprehensive review of YOLO architectures	Documents innovations from YOLOv1 to YOLOv8 and YOLO-NAS	General review; no queue-specific application
Wojke et al. [7]	2017	DeepSORT online multi-object tracking	Appearance and motion association reduce identity switches during occlusion	No explicit queue-position semantics
Ye et al. [8]	2022	Deep learning for person re-identification	Categorizes feature learning, metric learning, and ranking approaches	No integration with queue management
Padilla-Lopez et al. [6]	2015	Visual privacy protection methods	Categorizes anonymization, scrambling, and aggregation techniques	Not specific to queue or service-area monitoring
Shortle et al. [10]	2018	Foundational queueing theory	M/M/c and Little's Law enable wait-time calculation	No integration with CV inputs
Holt [11]	2004	Exponentially weighted moving average forecasting	Trend and level smoothing capture short-term dynamics	No queue-specific application
Shmueli & Koppius [9]	2011	Predictive analytics as a research paradigm	Empirical prediction supports decision-making	Not applied to live queue systems
Hyndman & Athanasopoulos [12]	2018	Forecasting principles and practice	Treats exponential smoothing and multi-horizon forecasting	Not domain-specific to service queues

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Curriculum Vitae

Appendices